KDD '26 · AI4SCIENCE TRACK

Condition-Gated Reasoning for Context-Dependent Biomedical Question Answering

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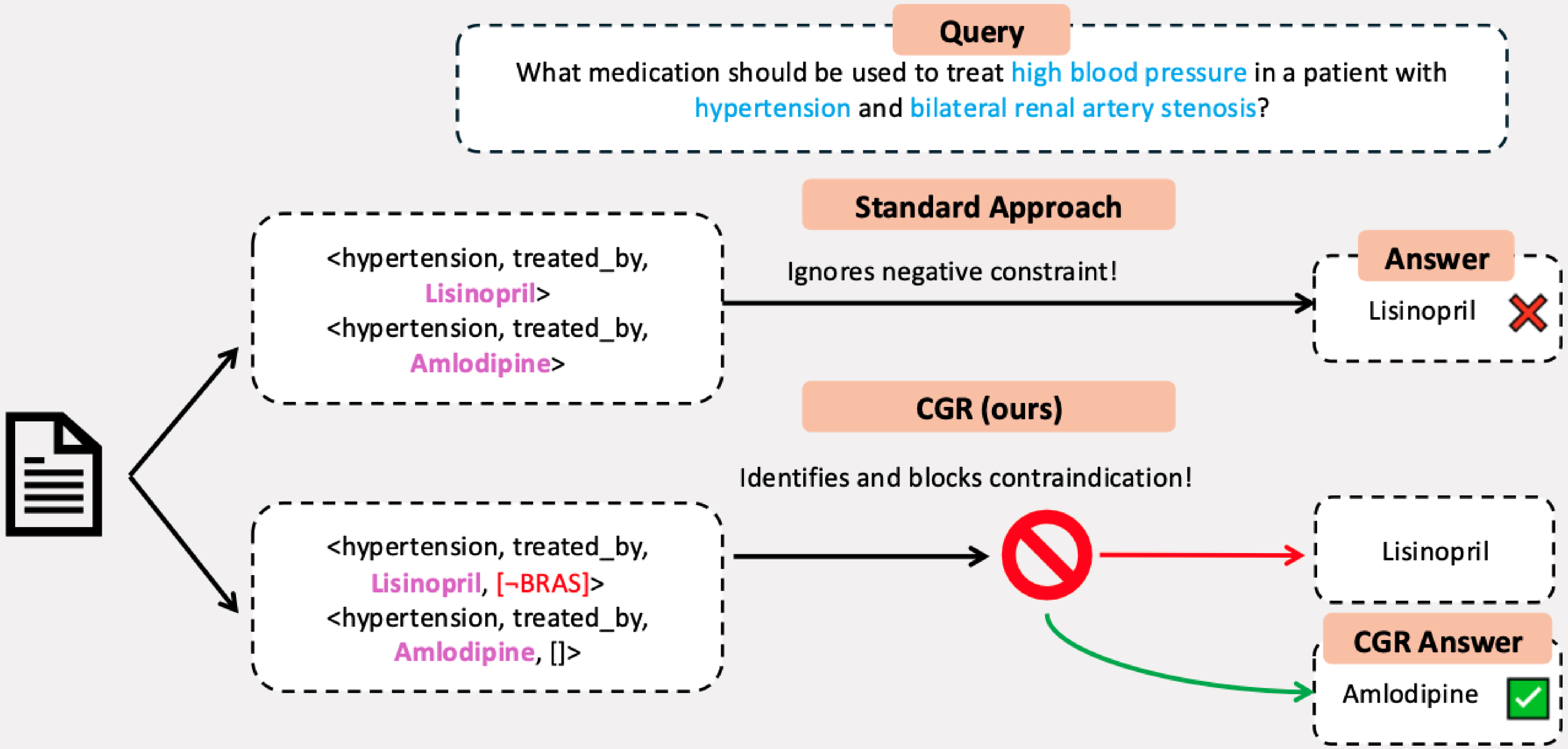
Motivating Example

Biomedical QA is not a fact-lookup task.

01 The answer depends on patient-specific factors — allergies, comorbidities, concurrent medications.

02 One question can have several correct answers, one per patient context.

03 A default treatment may be contraindicated: the best answer shifts with the constraint.



⟨lisinopril, treats, hypertension⟩ is true and **useless** without knowing the *exact* conditions about when it applies.

Two Gaps in Current Approaches

GAP 01 · REPRESENTATION

Triples have no slot for a condition

A graph of ⟨`entity, relation, entity`⟩ cannot state the circumstances under which the relation holds. Contraindications, patient-population restrictions and stage-specific guidelines are discarded at extraction time.

GAP 02 · EVALUATION

No benchmark tests answer revision

MedHopQA, BioASQ and PubMedQA measure factual recall and multi-hop composition. None of them measures whether an answer moves when a patient-specific constraint is added.

01 **CondMedQA** — a benchmark that isolates the behaviour.

02 **Condition-Gated Reasoning** — a method that produces it.

Talk Overview

01

The Benchmark

What makes a question conditional, and how we built 100 of them.

02

The Method

Conditions on edges, resolved once per query, enforced during traversal.

03

The Evidence

Four benchmarks, three foundation models, and what the gate is worth.

Defining Conditionality

01	**Modifier presence**	The question names a specific patient condition.
02	**General answer exists**	A standard answer applies if the modifier were absent.
03	**Answer divergence**	The conditional answer differs from the general one.
04	**Dual validity**	Both answers are correct — each in its own context.
05	**Causal dependence**	Strike the modifier and the general answer returns.

WORKED EXAMPLE

Which antidepressant for major depressive disorder is contraindicated in patients with bulimia?

MODIFIER

Bulimia nervosa

GENERAL ANSWER

Sertraline

CONDITIONAL ANSWER

Bupropion lowers the seizure threshold

Benchmark Composition

57%

Comorbidity & organ-based

Disease contraindications and organ dysfunction that force a drug substitution.

23%

Diagnostic modality

Imaging choice under radiation exposure and contrast contraindications.

16%

Special populations

Pregnancy safety, pediatric dosing, geriatric precautions.

4%

Drug–drug interactions

CYP enzyme interactions and pharmacogenomic variation.

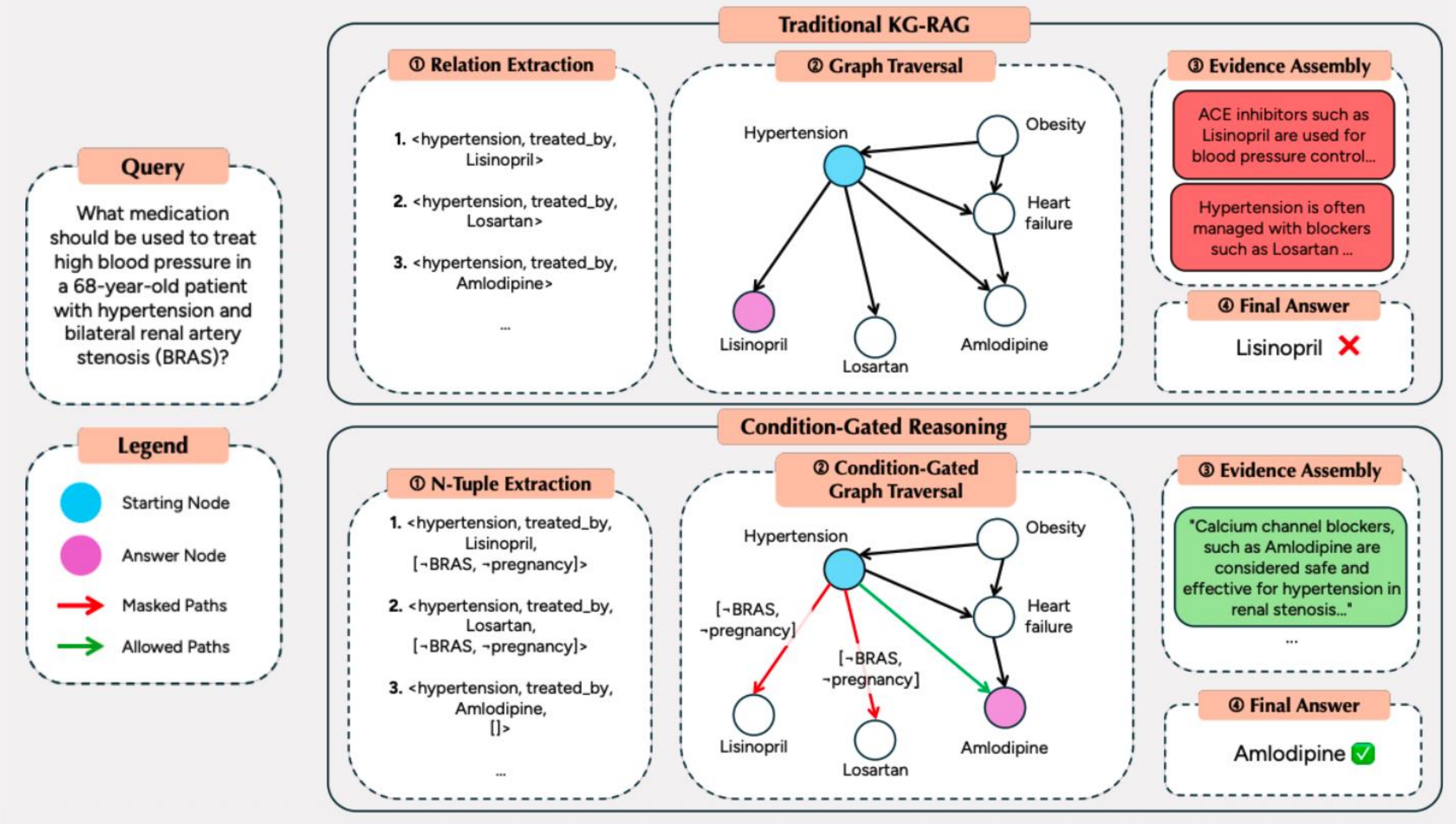
BUILT IN THREE STAGES

01 LLM candidate generation

02 Automated filtering

03 Manual expert verification

CGR vs. Traditional KG-RAG



TOP BAND

Every treatment edge is followed. A triple has no field for the BRAS, so lisinopril reaches the generator and becomes the answer.

BOTTOM BAND

The same edges now carry conditions. Two are masked, one survives — and only the surviving path's evidence reaches the generator.

Condition-Aware Knowledge Graph

N-TUPLE EXTRACTION

⟨u, r, v, C⟩

C is the list of conditions under which the relation holds or fails. In practice these are demographics, physiological status, comorbidities, disease stage and contraindication contexts.

ENTITY NORMALIZATION

UMLS Metathesaurus

Surface forms collapse onto canonical concepts — *heart attack* and *myocardial infarction* become one node, so conditions accumulate on the same edge.

GRAPH CONSTRUCTION

G = (V, E)

Every edge carries its conditions C and its source snippets S. Parallel edges between the same pair are allowed, so one drug can treat one disease under several distinct conditions.

EXTRACTED TUPLES

⟨metformin, treats, type_2_diabetes, **[¬severe_renal_impairment]** ⟩ ⟨misoprostol, contraindicated_in, wanted_pregnancies, **[pregnant_women]**⟩

Query Processing

QUERY PARSING

```
parse(q) = {K, C, N}
```

`K` entity keywords that seed traversal — drug, hypertension

`C` required and excluded conditions — in pregnancy

`N` negated entities to exclude — ACE inhibitors

Parser: Qwen2.5-14B-Instruct.

CONDITION EVALUATION

$$\texttt{eval}: C_G \times q \rightarrow \{T, F, \emptyset\}$$

`true` the query satisfies the condition

`false` the query violates it — the only blocking verdict

`null` unknown, and therefore not blocking

A single batched pass builds the lookup table L; every later edge test is O(1).

WORKED EXAMPLE — "WHAT ANTIBIOTIC FOR A 5-YEAR-OLD BOY WITH PNEUMONIA?"

"in children" → **true** · "in adults" → **false** · "renal impairment" → **null**

Condition-Gated Graph Traversal

EDGE GATING FUNCTION

$$G(e, L) = \prod_{c \in C(e)} \mathbb{1}[L(c) \neq \text{false}]$$

TRAVERSAL TERMINATES WHEN

Max depth d_{max} is reached · no outgoing edge passes the gate · the node is a leaf.

WALKTHROUGH — "MEDICATION FOR HYPERTENSION WITH BILATERAL RENAL ARTERY STENOSIS"

⟨hypertension, treated_by, lisinopril, [¬BRAS, ¬pregnancy]⟩	L(¬BRAS) = false	BLOCKED
⟨hypertension, treated_by, losartan, [¬BRAS]⟩	L(¬BRAS) = false	BLOCKED
⟨hypertension, treated_by, amlodipine, [ ]⟩	no conditions to violate	TRAVERSED

Blocked edges are removed from the search space, not ranked down — the answer generator never sees the contraindicated path.

Path Ranking and Answer Generation

PATH RANKING

$$\mathrm{score}(q, p) = \sum_{k \in K} \cos(\varphi(\mathrm{chain}(p)), \varphi(k))$$

Aggregate cosine similarity between the path's relation chain and each query keyword. Embeddings from MedEmbed-large-v0.1; the top N paths are kept.

EVIDENCE ASSEMBLY

$$E_q = \{(V_p, E_p, S_p, C_p)\}$$

Each retained path keeps its entity sequence, its source text snippets, and the conditions on every edge it traversed — so the generator can see why the path was admissible.

ANSWER GENERATION

$$A = \mathrm{LLM}(q \oplus P_q \oplus E_q)$$

A single generation call over the query, the ranked paths and their evidence. The backbone model is swappable — nothing in the gate is model-specific.

Cost. One batched LLM call for condition evaluation regardless of graph size, then $O(|V| + |E|)$ traversal with constant-time gate checks per edge.

Main Results

METHOD	CONDMEDQA EM	CONDMEDQA F1	MEDHOPQA EM	MEDHOPQA F1	MEDHOP (C) EM	MEDHOP (C) F1	BIOASQ-B EM	BIOASQ-B F1
Zero-Shot	22.0	41.3	20.5	47.9	28.3	53.5	13.0	40.2
CoT	40.0	52.8	49.5	57.6	50.0	59.4	25.6	47.8
HippoRAG2	46.0	67.6	45.0	60.8	48.6	72.8	22.6	43.6
MKRAG	60.0	71.1	51.3	65.6	62.9	79.3	28.8	55.4
RAG	44.0	52.4	51.3	62.6	45.7	55.0	31.8	54.8
StructRAG	62.0	71.2	71.8	79.6	65.7	78.6	31.8	55.1
MedRAG	57.0	70.5	75.8	84.4	74.3	**89.3**	35.4	55.5
CGR (ours)	**82.0**	**86.7**	**86.8**	**89.9**	**80.0**	85.4	**40.6**	**58.0**

+20.0 EM on CondMedQA · **+11.0 EM** on MedHopQA · leads on BioASQ, where nothing is conditional

Exact match and token-level F1 · post-retrieval setting over gold documents

Ablation: Condition Gating

+25.0

CondMedQA

EM, 57.0 → 82.0 — every question is conditional

+17.1

MedHopQA (Cond)

EM, 62.9 → 80.0 — the conditional slice

+14.3

MedHopQA

EM, 72.5 → 86.8 — mostly non-conditional

CONFIGURATION	CondMedQA EM	F1	MedHopQA EM	F1	MedHop (C) EM	F1
CGR without gating	57.0	60.2	72.5	76.8	62.9	73.5
CGR (full)	**82.0**	**86.7**	**86.8**	**89.9**	**80.0**	**85.4**

Extracting conditions is not sufficient. The gain comes from enforcing them ***during search***, and it concentrates where conditions actually bind.

Error Analysis

11 of 18 · 61%

Retrieval

The gold entity is in the graph and admissible, but a high-degree generic edge outranks the condition-appropriate path, so it misses the top-k.

5 of 18 · 28%

Extraction

The entity never entered the graph, or UMLS resolved it wrongly — `SSRI → SsrI endonuclease`.

2 of 18 · 11%

Reasoning

The evidence was right and the model answered the wrong question — naming the drug to replace instead of the replacement.

Ranking is purely semantic today — it does not know a path survived the gate. Both dominant categories are addressable without changing the gate.

Limitations and Future Work

Benchmark scope

100 questions makes CondMedQA a diagnostic instrument, not a training set. Broader coverage across condition types and clinical specialties is the next step.

Knowledge-source dependence

CGR inherits the errors, omissions and biases of its source documents. Extraction quality bounds everything downstream of it.

Ranking is condition-blind

Path ranking uses semantic similarity alone. Folding a condition-match signal into the score would address the dominant error category.

Entity normalization

UMLS CUI resolution can compound errors — `SSRI → SsrI endonuclease` . Auditing that pipeline is necessary work.

Future directions. Condition-aware path ranking, multi-modal knowledge graphs, privacy-preserving deployment, demographic bias auditing, and extension to other high-stakes domains.

Knowing *when* knowledge applies is a requirement.

BENCHMARK

100 questions where a patient-specific constraint changes the correct answer.

METHOD

Conditions ride on the edge; the gate enforces them during traversal.

RESULT

82.0 EM on CondMedQA (+20), 86.8 on MedHopQA (+11), stable across models.

NEXT

Condition-aware path ranking, broader condition coverage, bias auditing.

Questions?

PAPER

BENCHMARK

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